

Learner-State-Aware Generative User Interfaces: A Closed-Loop Control Architecture for Adaptive Learning Environments

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Abstract—Intelligent tutoring systems estimate what a learner knows but present a fixed interface, while generative user interfaces (GenUI) synthesize interfaces at runtime without any model of the learner. We propose a closed-loop architecture that couples the two. Bayesian Knowledge Tracing estimates mastery of a knowledge component; a discrete-time PID governor with anti-windup clamping and a filtered derivative converts the mastery deficit into a structural complexity budget; a fast policy selects discrete scaffolding affordances; and a foundation model emits a declarative JSON component tree that must satisfy the budget, with a deterministic fallback when schema or budget checks fail. A computable metric of interface complexity closes the loop between the budget and the rendered tree. We evaluate the architecture in silico on 1,000 synthetic learner trajectories drawn from three learner archetypes and replayed through four experimental arms. We report constraint adherence, interface jitter, recovery after error streaks and schema compliance **[TBD: headline numbers]**. We further show that standard BKT saturates after long success runs, so that a subsequent streak of errors leaves the interface unchanged, and evaluate forgetting and a CUSUM lapse detector as remedies. All claims concern algorithmic behavior; no learning gains are claimed.

Index Terms—generative user interfaces, Bayesian knowledge tracing, PID control, cognitive load, adaptive learning, simulation

I. INTRODUCTION

Intelligent tutoring systems (ITS) have long modeled learners precisely. Knowledge tracing maintains a probability that a learner has mastered each knowledge component [1], [2], and tutors built on it have been deployed at scale in mathematics [3], [4]. The interface of such systems, however, is authored in advance: the learner model selects the next problem, not the form in which the problem is presented.

Generative user interfaces take the opposite position. A foundation model produces a task-specific interface for each request [5], including interactive courseware [6]. These systems are open loop: the model does not observe the learner's state, and nothing bounds how complex the generated interface becomes.

Cognitive load theory explains why this matters. The same material can help a novice and hinder an expert (the expertise reversal effect) [7], [8], and guidance should fade as competence grows [9], [10]. An interface that adapts to the

learner therefore needs both a learner model and a mechanism that holds the interface to a complexity appropriate to that model, without the abrupt layout changes that make adaptive interfaces hard to use [11], [12].

This paper makes three contributions:

- 1) A closed-loop architecture in which a BKT mastery estimate drives a PID governor whose output is a structural complexity budget that a generative model must satisfy, with schema validation and a deterministic fallback.
- 2) A computable structural complexity metric over a declarative component tree for linear-equation tasks, built from features linked in prior work to cognitive load and visual complexity.
- 3) An in-silico benchmark of 1,000 synthetic learners across four arms and ablations, including an analysis of BKT saturation after success runs and two remedies.

II. RELATED WORK

A. Learner modeling

Bayesian Knowledge Tracing (BKT) models each knowledge component as a two-state hidden Markov model with prior, learning, guess and slip parameters [1]. Extensions individualize the prior [13] and the learning rate [14], add forgetting [15], [16], or contextualize guess and slip [17]. Its identifiability and degeneracy have been analyzed in detail [18]–[20]. Logistic models such as PFA [21], [22] and deep models [23] are alternatives; BKT remains competitive when extended [16] and is reviewed in [24], [25]. We use BKT because its state is a calibrated probability that can serve directly as a control variable.

B. Adaptive and generative interfaces

Adaptive hypermedia [26] and model-based interface generation [27] adapt presentation to users, but adaptation has costs: unpredictable changes reduce usability [11], [28], [29]. Recent GenUI systems let a large language model produce the interface itself [5], [6]. To our knowledge, none of these systems bounds the generated interface by a learner model.

C. Cognitive load and scaffolding

Cognitive load theory [30]–[32] attributes intrinsic load to element interactivity [33]. Worked examples reduce load for algebra novices [34] and should fade into problem solving [9], [10]; concrete representations should fade into symbolic ones [35]. For linear equations, the balance model supports early understanding but breaks down with negative quantities [36], [37], and learners hold persistent misconceptions about equality and operating on the unknown [38]–[41].

D. Control and constrained generation

PID control with anti-windup and derivative filtering is standard practice [42]–[44], and feedback control has been applied to computing systems [45] and to difficulty adjustment in games [46]. Constrained decoding guarantees well-formed model output [47]–[50], though schema-constrained output is not perfectly reliable in practice [51] and format restrictions can affect output quality [52].

III. ARCHITECTURE

The loop has four layers (Fig. [TBD: architecture figure]): a psychometric estimator, a complexity governor, a fast policy, and a generative plant whose output passes a validation gate before a deterministic client renders it.

A. Mastery estimation

After response $r_n \in \{0, 1\}$ the mastery estimate is updated by Bayes' rule and the learning transition [1]:

$$P(L_n | r_n=1) = \frac{P(L_n)(1 - P(S))}{P(L_n)(1 - P(S)) + (1 - P(L_n))P(G)}, \quad (1)$$

$$P(L_n | r_n=0) = \frac{P(L_n)P(S)}{P(L_n)P(S) + (1 - P(L_n))(1 - P(G))}, \quad (2)$$

$$P(L_{n+1}) = P(L_n | r_n)(1 - P(F)) + (1 - P(L_n | r_n))P(T), \quad (3)$$

where $P(F) = 0$ gives standard BKT and $P(F) > 0$ adds forgetting [16]. Default parameters are $P(L_0) = 0.15$, $P(T) = 0.12$, $P(G) = 0.20$, $P(S) = 0.08$. They are design values, not fitted; they satisfy the bounds $P(G) < 0.3$, $P(S) < 0.1$ [1] and the non-degeneracy conditions $P(G) + P(S) < 1$, $P(T) < 1 - P(S)/(1 - P(G))$ [19].

B. Complexity governor

The setpoint is the conventional mastery threshold $P^* = 0.95$ [1], [53]. The tracking error is normalized on each side of the setpoint so that deficit and surplus have equal authority:

$$e_n = \begin{cases} (P^* - P(L_n))/P^* & P(L_n) \leq P^* \\ (P^* - P(L_n))/(1 - P^*) & P(L_n) > P^* \end{cases}, \quad e_n \in [-1, 1]. \quad (4)$$

The control effort is

$$u_n = K_p e_n + K_i S_n + K_d D_n, \quad (5)$$

with the clamped integral $S_n = \text{clamp}(S_{n-1} + e_n, -S_{\max}, S_{\max})$ for anti-windup [44] and the first-order filtered derivative $D_n = \gamma(e_n - e_{n-1}) + (1 - \gamma)D_{n-1}$ [42]. An inverted logistic maps effort to the complexity budget,

$$M_I^* = (1 + e^{ku_n})^{-1}, \quad (6)$$

which is quantized into four levels $\ell_n = \min(4, 1 + \lfloor 4M_I^* \rfloor)$. A hysteresis deadband Δ_h commits a new budget only when it differs from the last committed budget by at least Δ_h .

Gain selection: Classical tuning rules [54]–[57] assume an identified plant response. In our benchmark the interface does not alter the simulated learner's responses (Section IV), so there is no plant response to identify. Instead, K_p , K_i , K_d , γ , $S_{\max}$ and Δ_h were selected by minimizing an ITAE-type tracking cost [58] plus a jitter penalty on a separate tuning cohort; all reported results use a disjoint test cohort. Selected values: [TBD: tuned gains from results/]. The logistic slope k is chosen so that the steady-state effort range maps to approximately [0.05, 0.95] [TBD: re-derive k for $P^* = 0.95$].

C. Fast policy

A fast policy maps $(P(L_n), M_I^*, \ell_n, \text{recent errors})$ to discrete affordances: scaffolding mode (worked example, guided scaffold, open symbolic), whether to show the balance-scale manipulative, whether hints are enabled, and an instructional density limit. The intended implementation is an open-weight decision model [59]; the benchmark uses a deterministic rule-based stand-in with the same interface, so no latency claims about that model are made. The split between a fast decider and a slow generator follows the System 1 / System 2 analogy [60].

D. Generative plant and validation gate

The plant ([TBD: Gemini model and configuration]) receives the budget, the policy decision and the equation and emits a JSON component tree. Nodes are drawn from a closed set of six components: CONCRETEBALANCESCALE, WORKEDSOLUTIONSTEP, SCAFFOLDEDOPERATIONPAD, SYMBOLICEQUATIONWORKSPACE, STEPPEDHINTACCORDION and CARTESIANVERIFICATIONPLANE. The client renders these components directly and never executes generated code. Each emission is parsed against the schema [61]; if parsing fails, or if $|M_I - M_I^*| > \varepsilon$ with $\varepsilon = 0.05$, a precompiled fallback tree nearest to the budget is rendered instead.

E. Structural complexity metric

For a tree with ρ interactive elements, abstraction level α and nesting depth δ ,

$$M_I = w_1 \frac{\rho - 1}{7} + w_2 \frac{\alpha - 1}{3} + w_3 \frac{\delta - 1}{3}, \quad \rho \in [1, 8], \alpha, \delta \in [1, 4]. \quad (7)$$

Each feature has a basis in prior work: element interactivity and choice count raise load [33], [62], [63] and display density is a classic complexity measure [64]; the abstraction ladder from balance scale to operation pad to symbolic workspace

TABLE I
LEARNER ARCHETYPES (UNIFORM PARAMETER RANGES).

Archetype	$P(L_0)$	$P(T)$	$P(G)$	$P(S)$
Struggling novice	.02–.08	.03–.07	.10–.20	.20–.30
Inconsistent guesser	.05–.20	.10–.20	.15–.40	.08–.30
Fast master	.15–.30	.30–.40	.15–.25	.01–.03

follows concreteness fading [35]; and deeper structures cost users more [65], [66]. Placing the Cartesian plane at the top of the ladder is our design choice. With no data to fit weights, we use equal weights $w_i = 1/3$, which are robust in that setting [67], and test sensitivity to the weights (Section V). M_I is a structural proxy; it has not been validated against measured cognitive load.

IV. METHOD

A. Synthetic learners

Simulated learners are an established way to test instructional designs before human studies [68], [69], and BKT-generated data have been used to evaluate mastery assessment [70], [71]. Each learner is a two-state HMM whose parameters are drawn uniformly from archetype ranges (Table I). The ranges for the novice and guesser archetypes exceed the tighter bounds of [1] by design to represent noisy learners, while respecting the non-degeneracy conditions of [19]. The cohort has 1,000 learners (300/400/300) with 40 items each. Responses are sampled once per learner and replayed unchanged through every arm (paired design). The interface therefore does not influence the simulated responses: the benchmark measures controller behavior, not learning.

B. Arms

(1) *Static*: a fixed level-2 interface. (2) *Unconstrained GenUI*: the plant chooses complexity from a noisy reading of mastery with no budget check. (3) *Rule-based*: step up after three consecutive correct responses and down after two consecutive errors, an up-down rule in the spirit of [72], [73]. (4) *Closed loop*: the full architecture. Ablations remove the derivative filter, remove anti-windup, and add a one-level slew limit.

C. Plant model in the benchmark

[TBD: If real Gemini rates are measured: describe them here. Otherwise:] The benchmark replaces the foundation model with a surrogate that compiles the closest admissible tree and injects faults: with probability [TBD: 0.15] it drifts off budget, and with probability [TBD: 0.03] it emits malformed output (truncated JSON, unknown component, out-of-range field, depth overflow, injected field). These rates are assumptions, not measurements.

D. Metrics

Constraint error $\Delta M = |M_I - M_I^*|$; schema compliance (valid trees over attempts); interface jitter $J = \frac{1}{N-1} \sum_n |M_I(n) - M_I(n-1)|$; recovery time after a five-error

TABLE II
MAIN RESULTS ON THE TEST COHORT (MEAN $\pm$ 95% CI).

Arm	ΔM	Jitter J	Schema ok	Fallback
Static	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Unconstrained GenUI	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Rule-based	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Closed loop	[TBD:]	[TBD:]	[TBD:]	[TBD:]

streak; and fallback rate. Jitter operationalizes the stability concern of [11]; the formula is ours.

V. RESULTS

[TBD: All numbers in this section come from the confirmed final benchmark run (commit, seed and CSV columns to be supplied by the engine).]

A. Ablations

[TBD: derivative filter, anti-windup, slew limit].

B. Sensitivity

[TBD: gains at $\pm 50\%$; metric weights: equal, 0.40/0.35/0.25, and each term dropped].

C. BKT saturation and lapse handling

Under correct responses the BKT estimate converges to a stable fixed point at one [19]. With the default parameters the estimate exceeds $1 - 10^{-6}$ after ten correct responses and five subsequent errors lower it only to about 0.98; after roughly forty correct responses it equals 1.0 in double precision and no further evidence can lower it. The interface therefore stays at its highest level through a streak of errors. We compare two remedies: forgetting, $P(F) > 0$ [15], [16], and a CUSUM detector [74], [75] on the log-likelihood ratio between “not known” and the tracker’s prediction, which resets the estimate when an alarm fires. [TBD: results on the forgetting cohort]. Weighting recent responses more heavily [76] is a third option we leave for future work.

VI. DISCUSSION AND LIMITATIONS

The benchmark tests whether the controller does what it is designed to do; it does not test whether learners benefit. Because simulated responses do not depend on the interface, the loop is closed only through the mastery estimate, not through the learner. M_I is a structural proxy that has not been validated against measured load, for which instruments exist [77]–[79]. BKT parameters are design values rather than fits to data such as ASSISTments [80]. Long error streaks may reflect misconceptions rather than slips [41], or a learner who will not reach mastery [81]; the controller does not distinguish these. [TBD: plant: surrogate vs. measured Gemini rates].

VII. CONCLUSION

[TBD: one paragraph after results are final].

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